THE RELATIONAL DATA MODEL

Formal Query Languages:
2. Relational Algebra Example Queries

Outline

1. Relational Algebra
2. Relational Algebra Example Queries
3. Relational Calculus

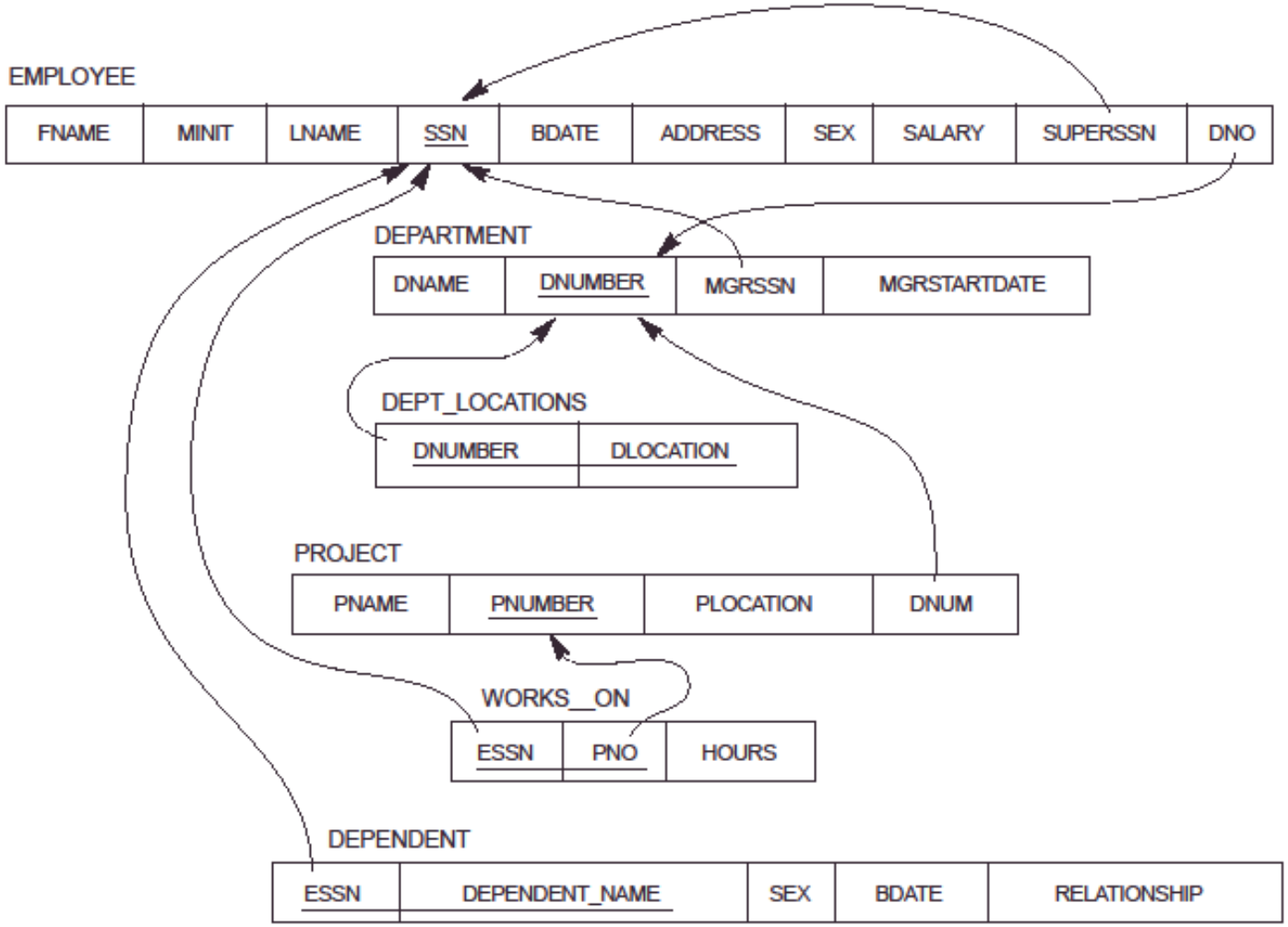
2.1. Remarks

- In general a query can be stated in different ways using the various operators.

- The presence of integrity constraints affects the way a query is expressed in relational algebra.

2.2. Example queries (1)

- *Retrieve the name and address of all employees who work for the 'Research' department.*



2.2. Example queries (2)

- *Retrieve the name and address of all employees who work for the 'Research' department.*

$$\text{RES_DEPT} \leftarrow \sigma_{\text{DNAME='Research'}}(\text{DEPARTMENT})$$

[DNAME, DNUMBER, MGRSSN, MGRSTARTDATE]

$$\text{RES_EMPS} \leftarrow \text{RES_DEPT} \bowtie_{\text{DNUMBER=DNO}} \text{EMPLOYEE}$$

[DNAME, DNUMBER, MGRSSN, MGRSTARTDATE, FNAME, MINIT, LNAME, SSN, …]

$$\text{RESULT} \leftarrow \Pi_{\text{FNAME, LNAME, ADDRESS}}(\text{RES_EMPS})$$

[FNAME, LNAME, ADDRESS]

2.2. Example queries (3)

- *Retrieve the name and address of all employees who work for the 'Research' department.*

  Alternative rewriting of the query (a natural join operator is used combined with an attribute renaming operator)

$$\text{RES_DEPT} \leftarrow \sigma_{\text{DNAME='Research'}}(\text{DEPARTMENT})$$

$$\text{RES_EMPS} \leftarrow \text{EMPLOYEE} * \rho_{\text{DNUMBER} \rightarrow \text{DNO}}(\text{RES_DEPT})$$

$$\text{RESULT} \leftarrow \Pi_{\text{FNAME, LNAME, ADDRESS}}(\text{RES_EMPS})$$

2.2. Example queries (4)

- *Retrieve the name and address of all employees who work for the 'Research' department.*

  Alternative rewriting of the query (the selection and join operations are reversed)

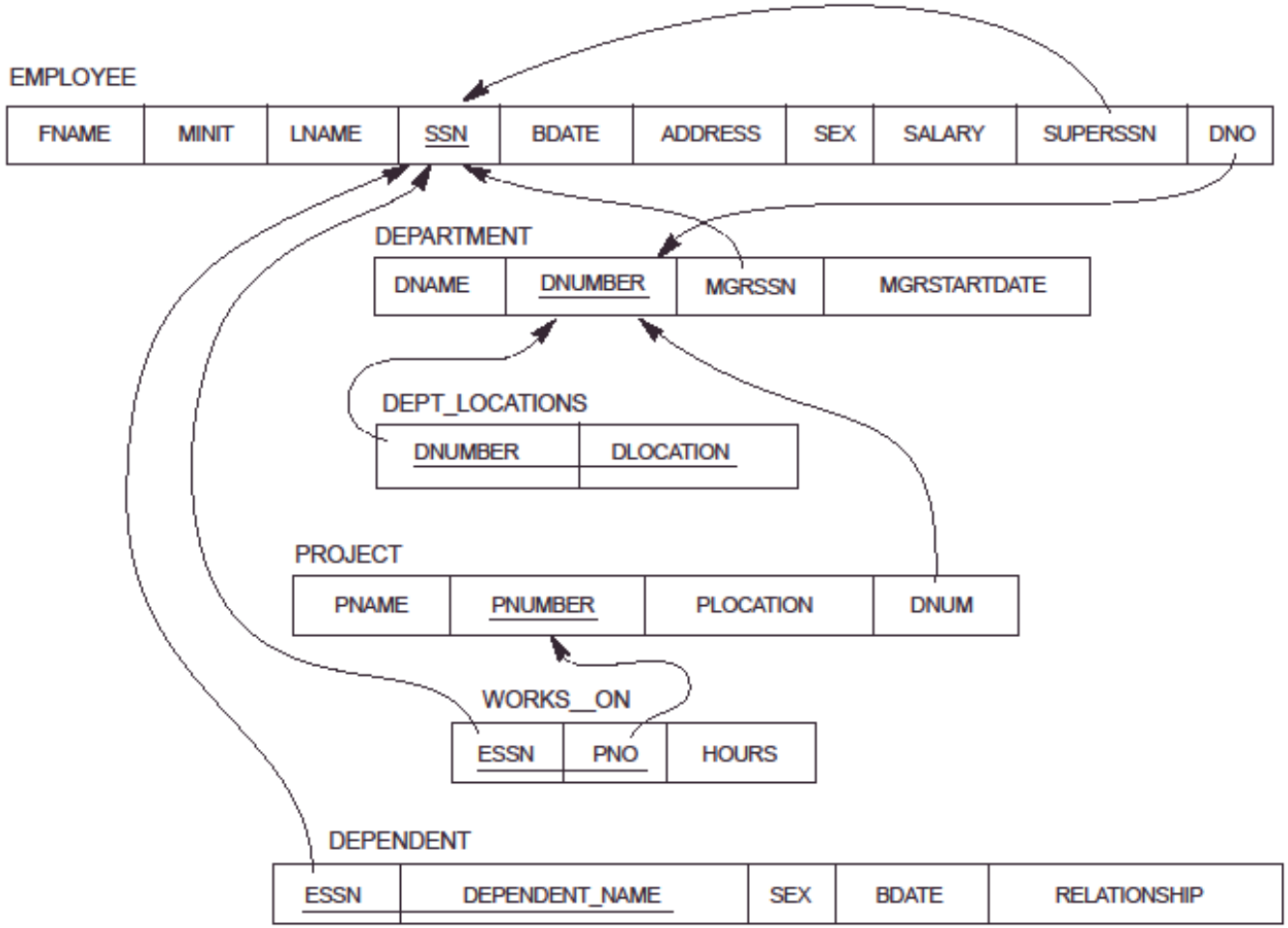
$$\text{EMP_DEPT} \leftarrow \text{EMPLOYEE} \bowtie_{\text{DNUMBER=DNO}} \text{DEPARTMENT}$$

$$\text{RES_EMPS_DEPTS} \leftarrow \sigma_{\text{DNAME='Research'}} (\text{EMP_DEPT})$$

$$\text{RESULT} \leftarrow \Pi_{\text{FNAME, LNAME, ADDRESS}}(\text{RES_EMPS_DEPTS})$$

2.2. Example queries (5)

- *For every project located in 'Stafford', list the project number, the controlling department number, and the department manager's last name, address and birthdate.*



2.2. Example queries (6)

- *For every project located in 'Stafford', list the project number, the controlling department number, and the department manager's last name, address and birthdate.*

$$\text{STAF_PROJS} \leftarrow \sigma_{\text{PLOCATION='Stafford'}}(\text{PROJECT})$$

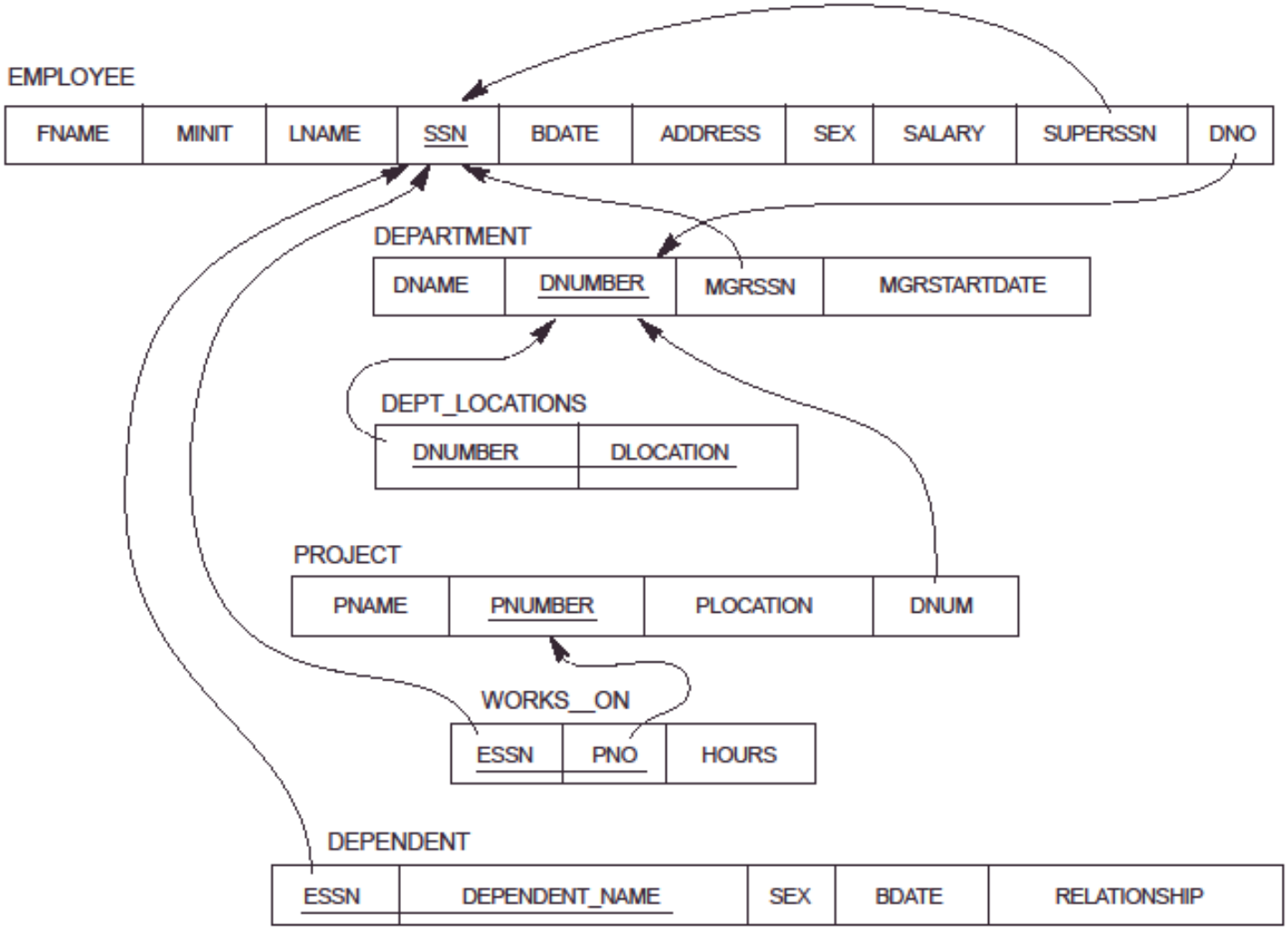
$$\text{CONT_DEPT} \leftarrow \text{STAF_PROJS} \bowtie_{\text{DNUM=DNUMBER}} \text{DEPARTMENT}$$

$$\text{PR_DEPT_MGR} \leftarrow \text{CONT_DEPT} \bowtie_{\text{MGRSSN=SSN}} \text{EMPLOYEE}$$

$$\text{RESULT} \leftarrow \Pi_{\text{PNUMBER, DNUM, LNAME, ADDRESS, BDATE}}(\text{PR_DEPT_MGR})$$

2.2. Example queries (7)

- *Find the names of employees who work on **all** the projects.*



2.2. Example queries (8)

- *Find the names of employees who work on **all** the projects.*

$$\underbrace{\text{ALL_PROJs}}_{[\text{PNUMBER}]} \leftarrow \Pi_{\text{PNUMBER}}\,(\text{PROJECT})$$

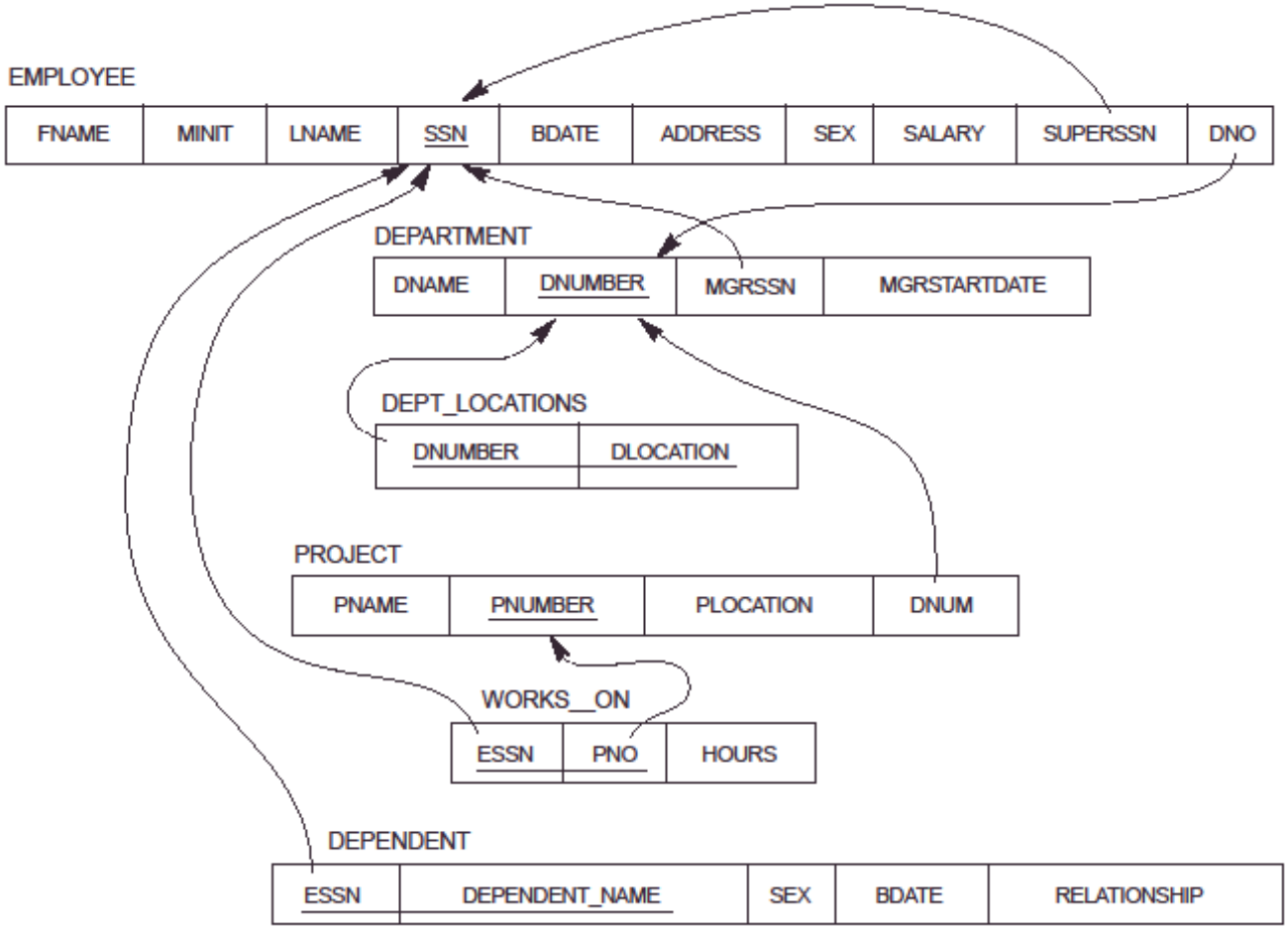
$$\underbrace{\text{EMP_PROJ}}_{[\text{ESSN, PNUMBER}]} \leftarrow \rho_{\text{PNO} \rightarrow \text{PNUMBER}}\,(\,\underbrace{\Pi_{\text{ESSN, PNO}}\,(\,\text{WORKS_ON}\,)}_{[\text{ESSN, PNO}]}\,)$$

$$\underbrace{\text{EMP_ALL_PROJs}}_{[\text{ESSN}]} \leftarrow \underbrace{\text{EMP_PROJ}}_{[\text{ESSN, PNUMBER}]} \div \underbrace{\text{ALL_PROJs}}_{[\text{PNUMBER}]}$$

$$\text{RES} \leftarrow \Pi_{\text{FNAME, LNAME}}\,(\,\text{EMP_ALL_PROJs} \rhd\!\lhd_{\text{ESSN=SSN}} \text{EMPLOYEE}\,)$$

2.2. Example queries (9)

- *Make a list of project numbers that involve an employee whose last name is 'Smith', either as a worker or as a manager of the department that controls the project.*



2.2. Example queries (10)

- *Make a list of project numbers that involve an employee whose last name is 'Smith', either as a worker or as a manager of the department that controls the project.*

$\underline{\text{SMITHS}} \leftarrow \rho_{SSN \rightarrow ESSN}(\Pi_{SSN}(\sigma_{LNAME='Smith'}(\text{EMPLOYEE})))$

[ESSN]

$\underline{\text{SMITH_WORKS}} \leftarrow \Pi_{PNO}(\text{WORKS_ON} * \text{SMITHS})$

[PNO]

$\text{SMITH_DPT} \leftarrow \Pi_{DUMBER}(\text{SMITHS} \rhd\lhd_{ESSN=MGRSSN} \text{DEPARTMENT})$

$\text{SMITH_DPT_R} \leftarrow \rho_{DNUMBER \rightarrow DNUM}(\text{SMITH_MGR})$

$\underline{\text{SMITH_PROJ}} \leftarrow \Pi_{PNUMBER}(\text{SMITH_DPT_R} * \text{PROJECT})$

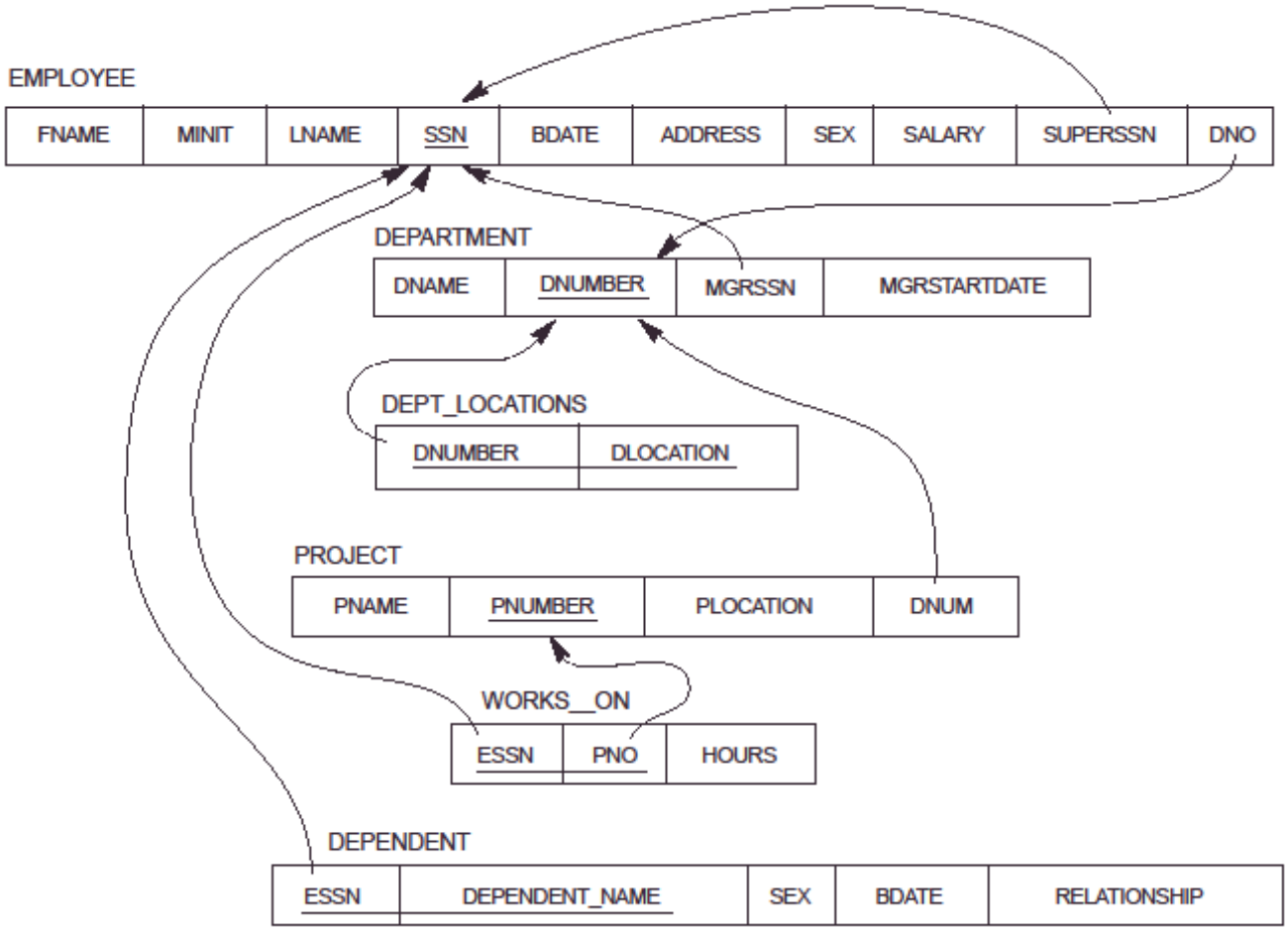
[PNUMBER]

$\underline{\text{RESULT}} \leftarrow \underline{\text{SMITH_WORKS}} \cup \underline{\text{SMITH_PROJ}}$

[PNO] [PNO] [PNUMBER]

2.2. Example queries (11)

- *List the names of all employees with two or more dependents.*



2.2. Example queries (12)

- *List the names of all employees with two or more dependents.*

$T1 \leftarrow {}_{ESSN}\mathcal{F}_{COUNT(DEPENDENT_NAME)}(DEPENDENT)$

[ESSN, COUNT(DEPENDENT_NAME)]

$T2 \leftarrow \rho_{ESSN \rightarrow SSN,\ COUNT(DEPENDENT_NAME) \rightarrow NO_OF_DEPS}(T1)$

[SSN, NO_OF_DEPS]

$T3 \leftarrow \sigma_{NO_OF_DEPS \geq 2}(T2)$

[SSN, NO_OF_DEPS]

$RESULT \leftarrow \Pi_{FNAME,\ LNAME}(\ T3 * EMPLOYEE\ )$

2.2. Example queries (13)

- *List the names of all employees with two or more dependents. (Alternative solution without aggregate functions.)*

$$\text{DEPEN} \leftarrow \Pi_{\text{ESSN, DEPENDENT_NAME}}(\text{DEPENDENT})$$

$$\text{DEPEN1} \leftarrow \rho_{\text{DEPENDENT_NAME} \rightarrow \text{DN1}}(\text{DEPEN})$$

[ESSN, DN1]

$$\text{DEPEN2} \leftarrow \rho_{\text{DEPENDENT_NAME} \rightarrow \text{DN2}}(\text{DEPEN})$$

[ESSN, DN2]

$$\text{T1} \leftarrow \text{DEPEN1} * \text{DEPEN2}$$

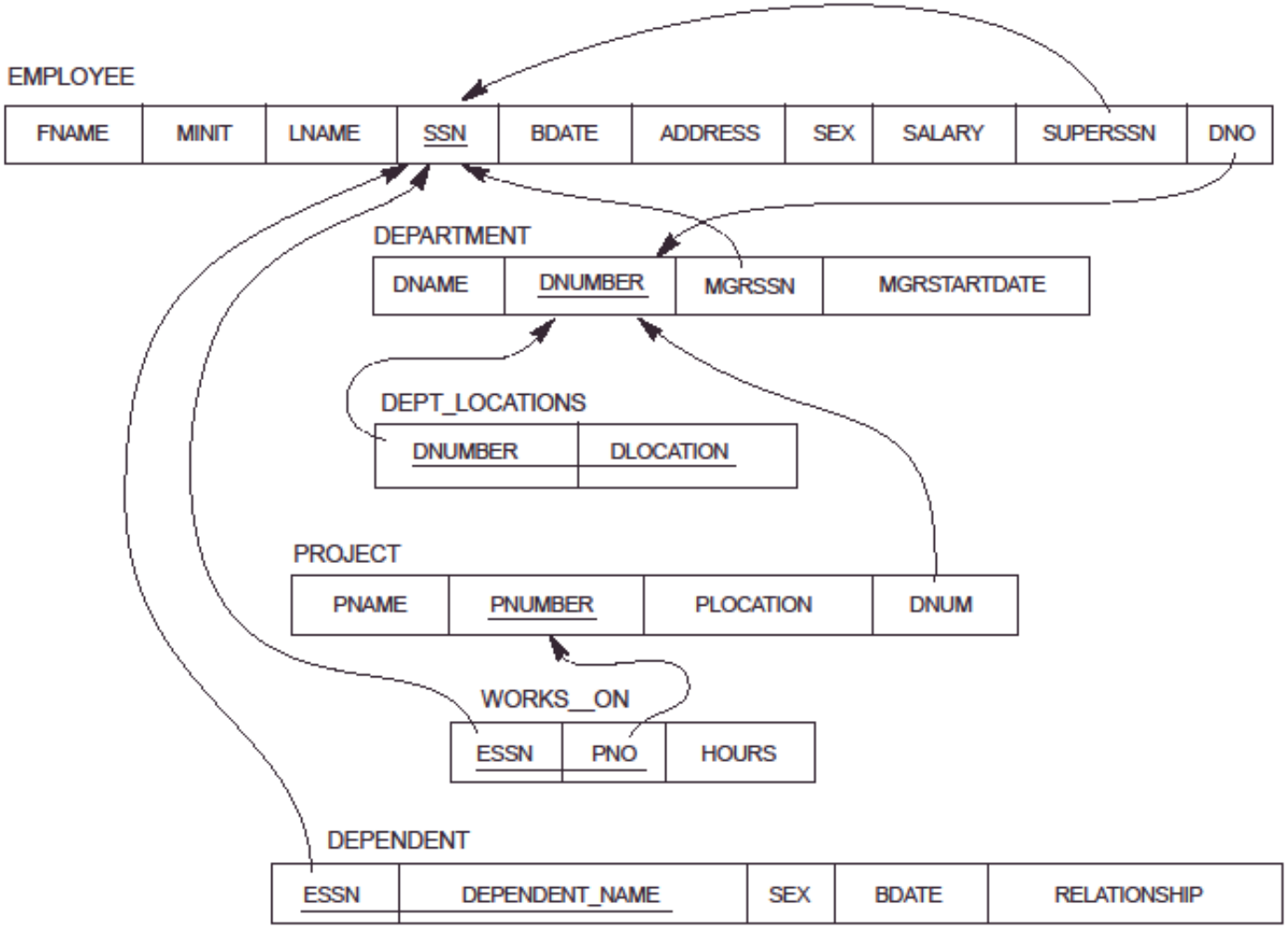
[ESSN, DN1, DN2]

$$\text{T2} \leftarrow \sigma_{\text{DN1} \neq \text{DN2}}(\text{T1})$$

$$\text{RESULT} \leftarrow \Pi_{\text{FNAME, LNAME}}(\text{T} \rhd\lhd_{\text{ESSN=SSN}} \text{EMPLOYEE})$$

2.2. Example queries (14)

- *Retrieve the names of employees who have no dependents.*



2.2. Example queries (15)

- *Retrieve the names of employees who have no dependents.*

$$\underline{\text{ALL_EMPS}} \leftarrow \Pi_{\text{SSN}}(\text{EMPLOYEE})$$

[SSN]

$$\underline{\text{EMPS_WITH_DEPS}} \leftarrow \Pi_{\text{ESSN}}(\text{DEPENDENT})$$

[ESSN]

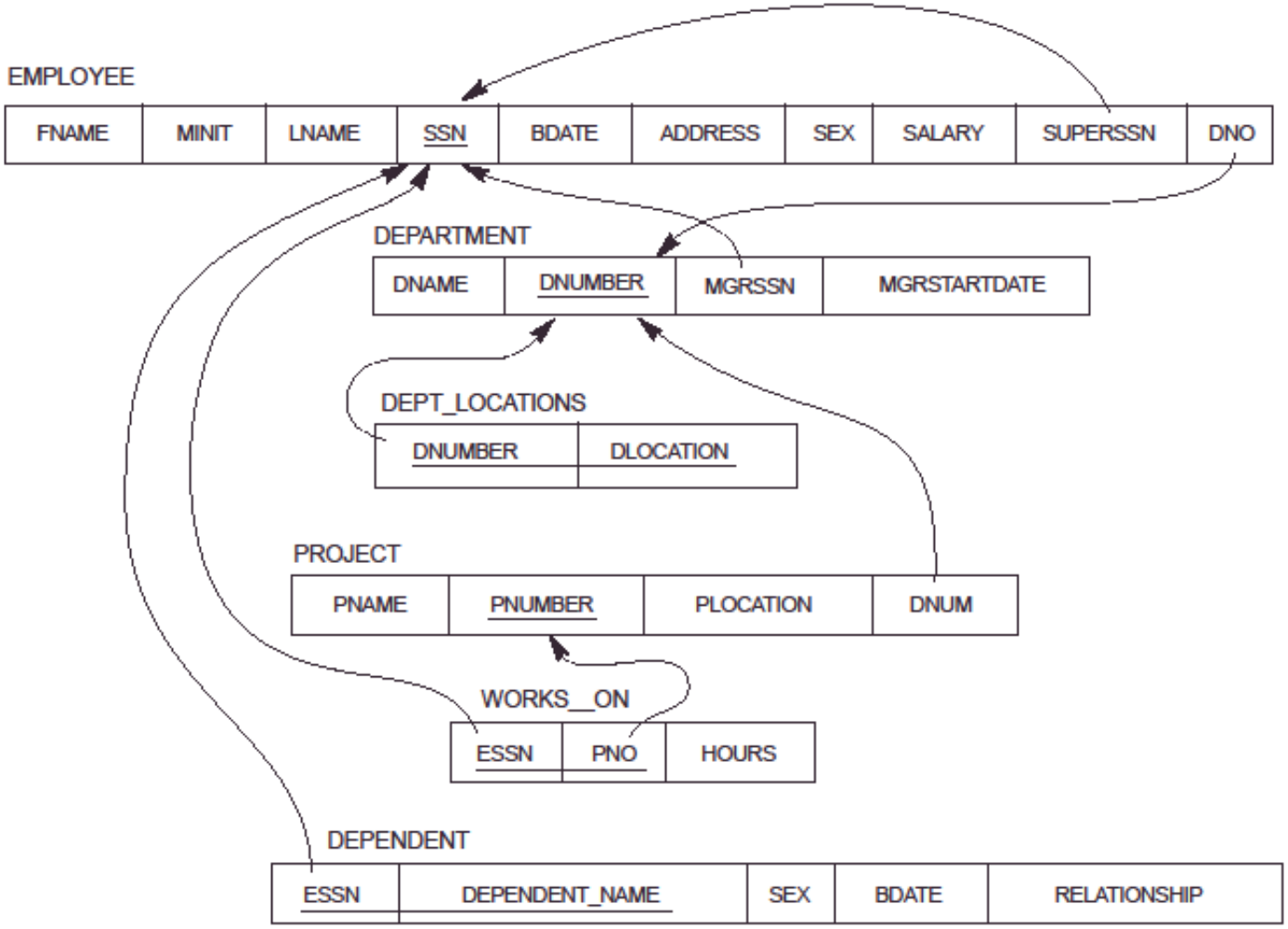
$$\underline{\text{EMPS_WITHOUT_DEPS}} \leftarrow \text{ALL_EMPS} - \text{EMPS_WITH_DEPS}$$

[SSN]

$$\text{RESULT} \leftarrow \Pi_{\text{FNAME, LNAME}}(\text{EMPS_WITHOUT_DEPS} * \text{EMPLOYEE})$$

2.2. Example queries (16)

- *List the names of managers who have at least one dependent.*



2.2. Example queries (17)

- *List the names of managers who have at least one dependent.*

$$\text{MGRS} \leftarrow \rho_{\text{MGRSSN} \rightarrow \text{SSN}} (\Pi_{\text{MGRSSN}} (\text{DEPARTMENT}))$$

$$\text{EMPS_WITH_DEPS} \leftarrow \rho_{\text{ESSN} \rightarrow \text{SSN}} (\Pi_{\text{ESSN}} (\text{DEPENDENT}))$$

$$\text{MGRS_WITH_DEPS} \leftarrow \text{MGRS} \cap \text{EMPS_WITH_DEPS}$$

$$\text{RESULT} \leftarrow \Pi_{\text{FNAME, LNAME}} (\text{MGRS_WITH_DEPS} * \text{EMPLOYEE})$$

2.2. Example queries (18)

- *List the names of managers who have at least one dependent (alternative solution).*

$$\text{MGRS_DEPS} \leftarrow \text{DEPENDENT} \rhd\lhd_{\text{ESSN= MGRSSN}} \text{DEPARTMENT}$$

$$\text{ESSN_MGRS_DEPS} \leftarrow \Pi_{\text{ESSN}} (\text{MGRS_DEPS})$$

$$\text{RESULT} \leftarrow \Pi_{\text{FNAME, LNAME}} (\text{ESSN_MGRS_DEPS} \rhd\lhd_{\text{ESSN= SSN}} \text{EMPLOYEE})$$

2.3. Query optimization

- All DBMSs have a *query optimizer*.
- A relational algebra query may have *many equivalent expressions* and some of them may be *much more easier to evaluate (compute)*.
- One of the tasks of a query optimizer is to generate *alternative equivalent expressions* of a query in order to find *one that is more efficient to evaluate*.